Relationship between Mammalian Brain Size and Other Biological Characteristics

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Abstract

Evolutionary biologists are keenly interested in what characteristics of a species determine their evolutionary pathway. A variable that is of particular interest to these biologists is brain size as investigations have shown plays a large role in the evolution of species. One might expect that bigger brains are better and indeed scientists claim that mammals with large brains tend to have more chance of surviving.

This project investigates the relationships between brain size and other biological characteristics among 96 species of mammals. Using data on average brain weight, body weight, gestation length, and litter size, the study explores how these variables are interrelated and what they reveal about mammalian evolution. Through exploratory data analysis and statistical modelling, the strength of association between brain weight and the other variables is assessed.

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1 Introduction

Understanding the factors that influence the evolution of brain size in mammals is a central question in evolutionary biology. Brain size has long been regarded as a key indicator of cognitive ability, behavioural complexity, and adaptive success among species. Larger brains are often associated with enhanced problem-solving capacity, complex social structures, and greater adaptability to changing environments. Consequently, biologists have sought to determine which physiological and reproductive characteristics are most closely related to brain development.

In this study, data from 96 mammalian species are analyzed to explore possible associations between brain weight and three other biological variables: body weight, gestation length, and litter size. These variables represent essential aspects of a species' life-history strategy—body weight reflects metabolic and energetic constraints, gestation length indicates parental investment and developmental time, while litter size reflects reproductive strategy and offspring survival trade-offs. Examining how these traits vary with brain size can provide valuable insight into evolutionary patterns and adaptive strategies across species.

The dataset, originally compiled by Ramsey and Schafer (1997) in The Statistical Sleuth, includes average values for each variable across multiple species. The analysis focuses on exploring relationships between variables, and interpreting the biological implications of observed patterns. By assessing how brain size relates to body size, reproductive investment, and offspring quantity, this study aims to contribute to a deeper understanding of the evolutionary pathways that have shaped mammalian diversity.

2 Data Description

The dataset used in this study was compiled by Ramsey and Schafer (1997) and contains information on 96 species of mammals. For each species, four key characteristics were recorded:

- **Brain Weight (g):** Average brain weight of the species in grams.
- **Body Weight (kg):** Average body weight of the species in kilograms.
- **Gestation Period (days):** Average length of pregnancy in days.
- **Litter Size:** Average number of offspring per litter.

These characteristics were selected because previous research suggests that brain size in mammals is influenced by body size, reproductive traits, and life history parameters.

The entire dataset is provided at the Appendix A.

2.1 Summary Statistics

The table below represents the summary statistics for each variable in the dataset.

Variable	Mean	Median	Minimum	Maximum	Std. Dev.
Brain weight	218.97698	74.0	0.45	4480.0	506.96801
Body weight	108.32775	8.89999	0.017	2800.0	329.41140
Gestation period	151.26041	133.5	16.0	655.0	108.06608
Litter size	2.31041	1.2	1.0	8.0	1.74702

Table 2.1: Summary statistics of each variable

3 Modelling Approach

The analysis was carried out in two distinct stages to gradually understand how different biological characteristics of mammals relate to brain size. The guiding idea was to start with simple relationships involving a single factor and then build towards a combined model that reflects biological complexity more realistically.

3.1 Stage 1: Independent Univariate Models

In the first stage, each explanatory variable was studied individually with brain weight. The goal was to examine whether *Body weight*, *Gestation length*, and *Litter size* are independently associated with *Brain weight*.
For each case, we considered a power–law type model of the form:

$$\text{Brain} = k \cdot (\text{Factor})^{\alpha} \varepsilon,$$

where k and α are constants to be estimated.
Three separate models were fitted:

(i) $\text{Brain} = k_1(\text{Body})^{\alpha_1}\varepsilon$

(ii) $\text{Brain} = k_2(\text{Gestation})^{\alpha_2}\varepsilon$

(iii) $\text{Brain} = k_3(\text{Litter})^{\alpha_3}\varepsilon$

Since the relationship is multiplicative and non-linear, we take the logarithm of both sides (using base 10) to obtain a linear equation:

$$\log(\text{Brain}) = \log(k) + \alpha \log(\text{Factor}) + \log(\varepsilon).$$

This transformation allows the model parameters to be estimated using straight-line fitting, where α represents the slope and $\log(k)$ the intercept. Plots of log(Brain) against log(Body), log(Gestation), and log(Litter) were constructed, and the best-fitting line in each case was determined.

3.2 Stage 2: Combined Model and Bio–Complexity Score

While each factor individually gives partial insight, real biological systems involve multiple interacting characteristics. In the second stage, these influences were combined into a unified power–law model:

$$\text{Brain} = k \cdot (\text{Body})^{\alpha} \, (\text{Gestation})^{\beta} \, (\text{Litter})^{\gamma} \varepsilon.$$

Taking logarithms again yields a linear additive form:

$$\log(\text{Brain}) = \log(k) + \alpha \log(\text{Body}) + \beta \log(\text{Gestation}) + \gamma \log(\text{Litter}) + \log(\varepsilon).$$

In class, we have been taught how to choose the best-fitting line in case of a simple model. Generalizing that procedure, we consider a composite **Bio–complexity Score** to capture the combined effect of the three predictors:

$$\text{Score} = \hat{\alpha} \log(\text{Body}) + \hat{\beta} \log(\text{Gestation}) + \hat{\gamma} \log(\text{Litter}) + \hat{\delta},$$

where $\hat{\alpha}, \hat{\beta}, \hat{\gamma}$ and $\hat{\delta}$ are chosen to minimise the quantity

$$S = \sum_{i=1}^{n} \left[\log(\text{Brain}_i) - \text{Score}_i\right]^2.$$

So, the values of $\hat{\alpha}, \hat{\beta}, \hat{\gamma}$ and $\hat{\delta}$ can be obtained by solving the equations:

$$\frac{\partial S}{\partial \hat{\alpha}} = \frac{\partial S}{\partial \hat{\beta}} = \frac{\partial S}{\partial \hat{\gamma}} = \frac{\partial S}{\partial \hat{\delta}} = 0,$$

A simple linear model between log(Brain) and Score was then fitted and the coefficient of determination was calculated.

Thus, Stage 2 integrates the individual effects observed earlier into a holistic model linking

brain weight to multiple biological traits, interpreted through the lens of evolutionary and reproductive complexity.

The frequency distribution of log(Brain) and the scatter plot matrix of the variables is shown below.

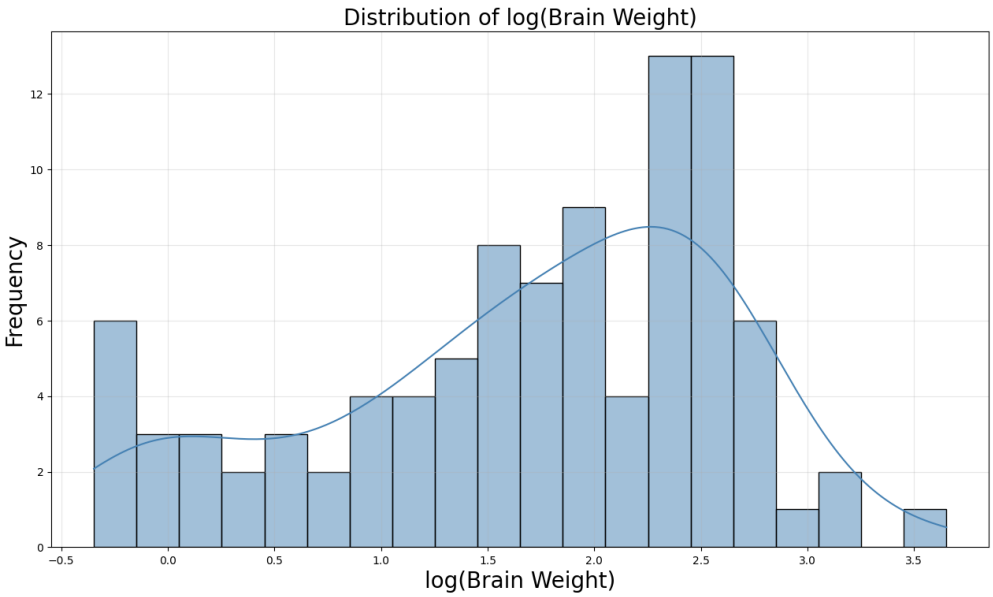

Figure 3.1: Frequency distribution of brain weight

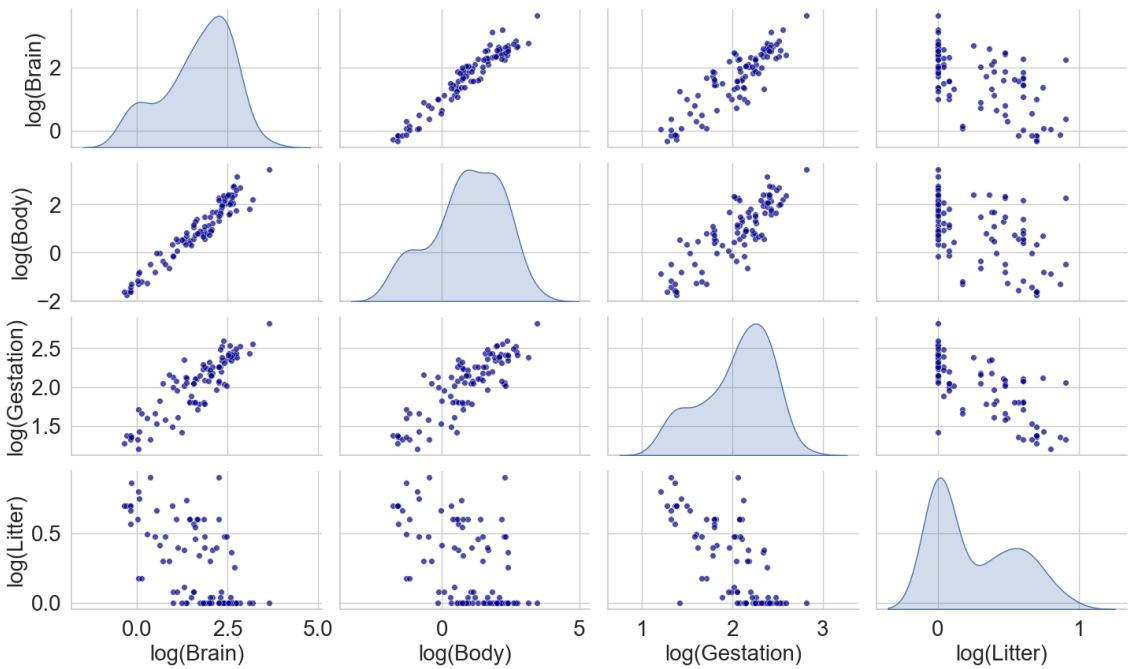

Figure 3.2: Scatter plot of the variables in the model

3.3 Model Assumptions

Before fitting the regression models in both stages, certain fundamental assumptions were made to ensure the validity of the results. These assumptions form the basis of the least

squares estimation method and determine whether the derived relationships between brain weight and the explanatory variables are statistically reliable and interpretable. The main assumptions are described below:

1. **Linearity:** It is assumed that, after taking logarithmic transformations, the relationship between log(Brain) and each predictor variable becomes linear.

2. **Homoscedasticity:** The variance of the residuals (errors) is assumed to remain constant across all fitted values.

3. **Normality of Errors:** It is assumed that the residuals follow an approximately normal distribution with mean zero.

4. **No Perfect Multicollinearity (for Stage 2):** In the multiple regression stage, it is assumed that the predictors log(Body), log(Gestation), and log(Litter) are not perfectly correlated with each other.

These assumptions were verified using diagnostic plots in residual analysis section (discussed later).

4 Result/Analysis of Stage 1

In this stage, we examine how each biological factor individually influences the size of the mammalian brain. To study the power–law relation between the variables, we assume models of the form

$$\text{Brain} = k(\text{Factor})^{\alpha}\varepsilon,$$

where k and α are constants to be determined from data. Taking the logarithm of both sides gives

$$\log(\text{Brain}) = \log(k) + \alpha \log(\text{Factor}) + \log(\varepsilon),$$

which is a linear relation between log(Brain) and log(Factor). Thus, α and $\log(k)$ can be estimated using simple linear regression.
The analysis was performed separately for the three biological factors: body weight, gestation period, and litter size. The fitted models and corresponding coefficients are summarized below.

$$\textbf{Model 1:} \quad \log(\widehat{\text{Brain}}) = 1.0129 + 0.7192 \log(\text{Body})$$
$$\textbf{Model 2:} \quad \log(\widehat{\text{Brain}}) = -2.8945 + 2.2340 \log(\text{Gestation})$$
$$\textbf{Model 3:} \quad \log(\widehat{\text{Brain}}) = 2.2189 - 2.0827 \log(\text{Litter})$$

Model	**Estimated values**		R^2
	$\hat{\alpha}$ **(slope)**	$\log(\hat{k})$ **(intercept)**	
Model 1	0.719	1.013	0.930
Model 2	2.234	−2.894	0.794
Model 3	−2.083	2.219	0.413

Table 4.1: Results of Stage 1: Individual Power–law Fits

4.1 Graphical Analysis

Scatter plots of log(Brain) versus log(Body), log(Gestation), and log(Litter) were drawn along with their fitted regression lines.

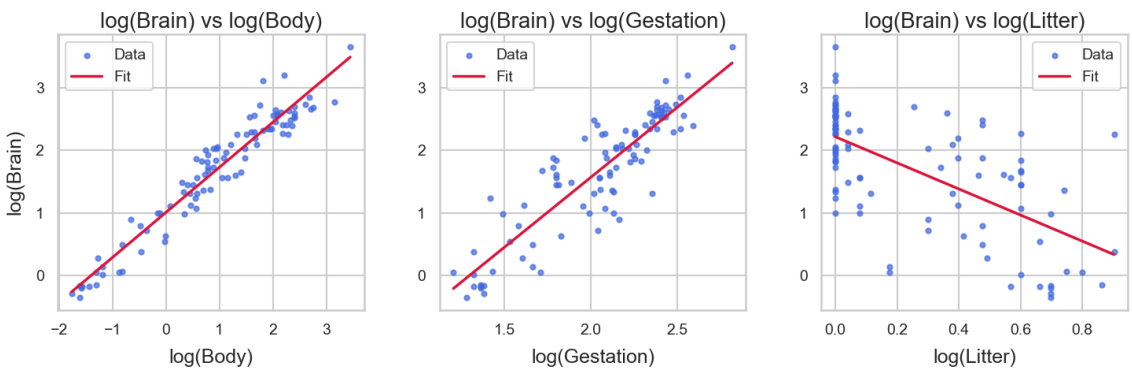

Figure 4.1: Plot showing best fitting line for each Model

All three relationships exhibit approximate linearity, confirming the suitability of the power–law model.

4.2 Inference

From Table 4.1, we observe that:

- The relationship between **brain weight and body weight** is the strongest, with $R^2 = 0.93$, indicating that nearly 93% of the variation in brain size can be explained by differences in body weight.
- The **gestation period** also shows a strong positive association ($\alpha = 2.23$, $R^2 = 0.79$), suggesting that mammals with longer pregnancies tend to have larger brains.
- The **litter size** exhibits a negative relationship ($\alpha = -2.08$), implying that mammals producing larger litters generally have smaller brains.

Overall, Stage 1 analysis suggests that mammals with larger bodies and longer gestation periods tend to have larger brains, while species with larger litter sizes generally have

smaller brains. This finding aligns with biological expectations, as brain development requires greater maternal investment and prolonged gestation, which are less compatible with large litter production.

5 Result/Analysis of Stage 2

After investigating the individual power-law relationships in Stage 1, the next step is to test a more comprehensive model that simultaneously incorporates all three biological factors. Biologists often expect that a mammal's brain size may depend not on a single variable alone but on a combination of its body size, gestation period, and litter size. Hence, we consider the multiplicative model:

$$\text{Brain} = k \times (\text{Body})^{\alpha} \times (\text{Gestation})^{\beta} \times (\text{Litter})^{\gamma} \times \varepsilon,$$

where k is a proportionality constant, and the exponents α, β, and γ represent the relative sensitivity of brain size to each factor. Taking logarithm on both sides give

$$\log(\text{Brain}) = \alpha \log(\text{Body}) + \beta \log(\text{Gestation}) + \gamma \log(\text{Litter}) + \delta + \log(\varepsilon),$$

where $\delta = \log(k)$. The estimates of α, β, γ and δ obtained by following the procedure as described in section 3.2 are 0.575, 0.418, -0.310 and 0.371 respectively.
The model achieved an R^2 value of approximately 0.954, indicating that about 95.4% of the variation in log-transformed brain weight is explained by the combined predictors.

5.1 Graphical Analysis

A scatter plot (Figure 5.1) of log(Brain) against the combined predictor

$$\text{Score} = 0.575 \log(\text{Body}) + 0.418 \log(\text{Gestation}) - 0.310 \log(\text{Litter}) + 0.371$$

shows a strong linear trend, further supporting the adequacy of the model. The regression line fits the observed data closely, with minimal deviation, confirming the suitability of the power-law formulation.

5.2 Inference

Although the combined model in Stage 2 achieves a slightly higher coefficient of determination ($R^2 = 0.954$) compared to the single-variable model using body weight alone ($R^2 = 0.930$), the improvement is relatively modest. This can be understood by examining the correlations among the explanatory variables.

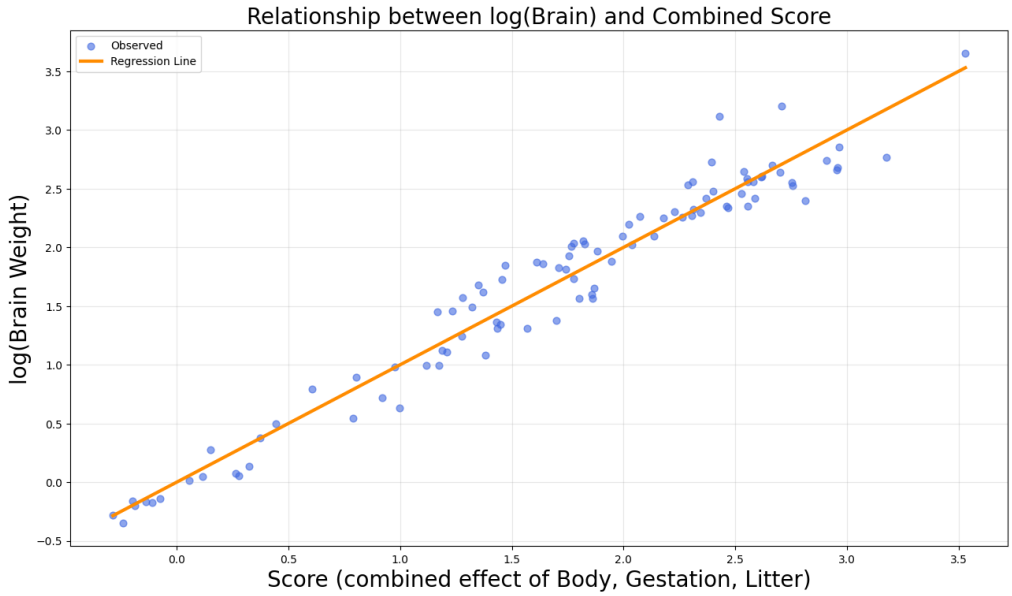

Figure 5.1: Plot showing best fitting line for the Model

	log(Brain)	**log(Body)**	**log(Gestation)**	**log(Litter)**
log(Brain)	1.000	0.964	0.891	−0.643
log(Body)	0.964	1.000	0.846	−0.546
log(Gestation)	0.891	0.846	1.000	−0.758
log(Litter)	−0.643	−0.546	−0.758	1.000

Table 5.1: Correlation matrix among logarithmic variables.

The correlation analysis reveals that the predictors—body weight, gestation period, and litter size—are themselves strongly interrelated. Body weight and gestation length are highly positively correlated ($r = 0.846$), while both show strong negative correlations with litter size ($r = -0.546$ and $r = -0.758$, respectively). Consequently, these predictors provide overlapping information about the biological structure underlying brain size. Because body weight already explains a major portion of the variance in brain weight ($r = 0.964$), adding gestation length and litter size contributes only marginal additional explanatory power. In evolutionary terms, this suggests that body size, gestation duration, and reproductive strategy are not independent traits but are all components of a shared life-history pattern: **larger mammals tend to have longer developmental periods and smaller litters**. Hence, the limited increase in R^2 from 0.93 to 0.95 primarily reflects the strong multicollinearity among these biological variables rather than a lack of biological association.

6 Residual Analysis

The residual plots and histograms for both Stage 1 and Stage 2 models are shown in Fig. 6.1 and Fig. 6.2, respectively.

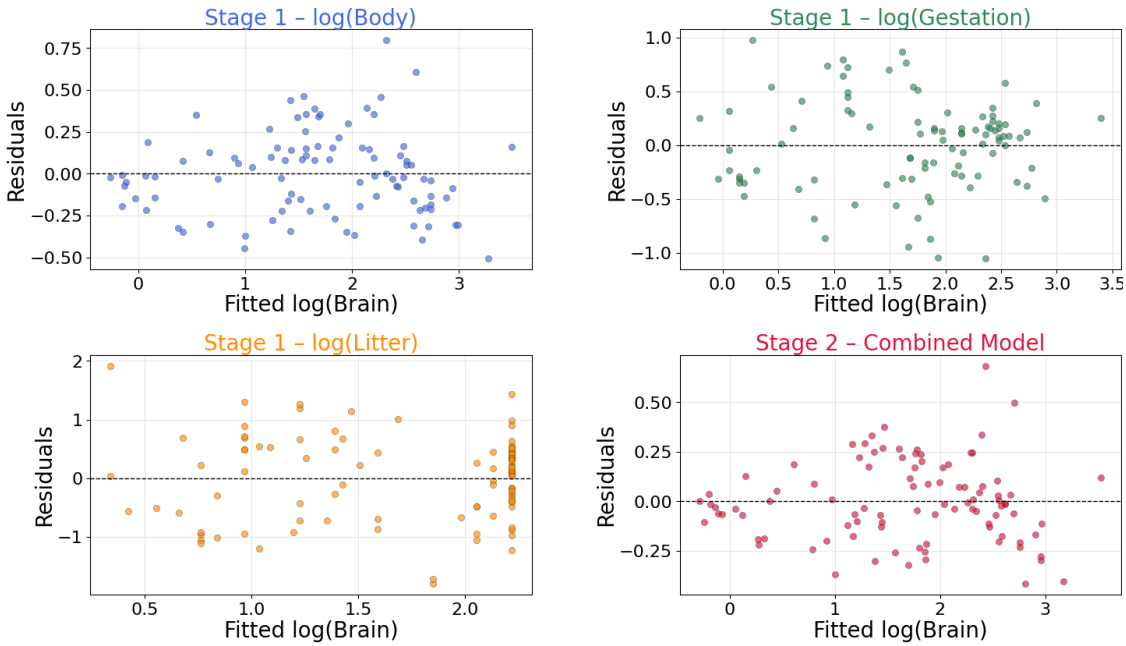

Figure 6.1: Residual plots for Stage 1 and Stage 2 models.

6.1 Interpretation

The residual for each observation is given by

$$e_i = y_i - \hat{y}_i,$$

where y_i denotes the observed value of log(Brain) and $\hat{y}_i$ denotes the predicted value from the fitted model.

- The residuals for the **Stage 1 models** (log Body, log Gestation, and log Litter) appear roughly centered around zero, indicating that the *linearity assumption* is reasonably satisfied.
- The residuals of the **Stage 2 combined model** show a smaller spread compared to individual Stage 1 models, suggesting that including multiple predictors has reduced the unexplained variance.
- The **histograms of residuals** show approximate symmetry about zero, which supports the *normality assumption* of residuals. However, slight deviations—particularly in the log Litter model—suggest some heteroscedasticity or skewness for smaller species.

- Overall, the residual patterns indicate that the regression assumptions hold approximately true, and the models can be considered reliable for interpretation.

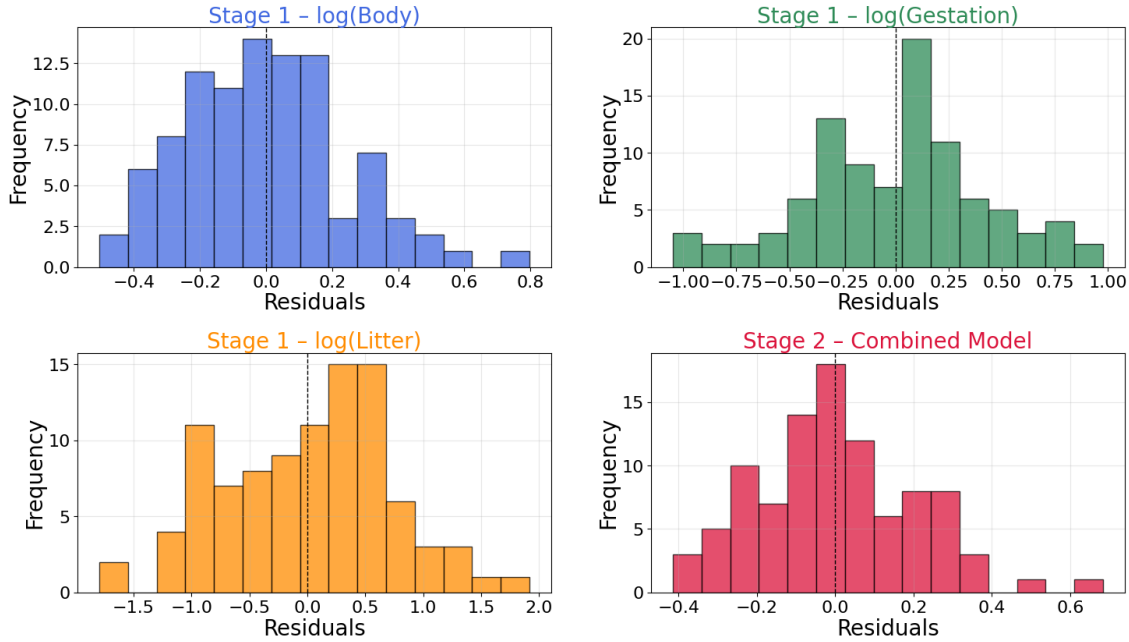

Figure 6.2: Histograms of residuals for Stage 1 and Stage 2 models.

7 Conclusion

The study aimed to investigate the relationship between brain weight and key biological characteristics—body weight, gestation length, and litter size—across a wide range of mammalian species. Using logarithmic transformations, the relationships were modelled as power laws of the general form

$$\text{Brain} = k \cdot (\text{Body})^{\alpha} \cdot (\text{Gestation})^{\beta} \cdot (\text{Litter})^{\gamma}.$$

Analysis from Stage 1 revealed strong individual correlations between brain weight and body weight ($R^2 \approx 0.93$), with weaker yet meaningful associations with gestation length ($R^2 \approx 0.79$) and litter size ($R^2 \approx 0.41$). When all three factors were combined in Stage 2, the explanatory power increased only slightly ($R^2 \approx 0.95$), suggesting that body weight alone already accounts for most of the variation in brain size among mammals. This modest gain in explanatory strength, despite the inclusion of multiple variables, indicates substantial interdependence between the predictors—particularly between body weight and gestation length—reflecting the biological coupling of size, development time, and reproduction in mammals.

The resulting model,

$$\log_{10}(\widehat{\text{Brain}}) = 1.013 + 0.719 \log_{10}(\text{Body}),$$

or equivalently,

$$\boxed{\widehat{\text{Brain}} = 10.304 \times (\text{Body})^{0.719}},$$

provides a simple yet robust formula for predicting average brain weight from body weight. It captures the principle that brain size does not increase linearly with body size: larger mammals tend to have disproportionately smaller brains relative to their bodies—a hallmark of evolutionary scaling.

Interval Estimation. The allometric relation

$$\text{Brain} = k\,(\text{Body})^{\alpha}\varepsilon$$

can be linearized by taking logarithms on both sides:

$$\log(\text{Brain}) = \log k + \alpha \log(\text{Body}) + \log(\varepsilon),$$

which corresponds to a simple linear regression model with intercept $\log k$ and slope α. Hence, the 95% confidence intervals for $\log k$ and α are obtained as:

$$\log k \in [0.950,\ 1.076], \qquad \alpha \in [0.678,\ 0.760].$$

Exponentiating the interval for $\log k$ gives the corresponding interval for k:

$$k \in \left[10^{0.950},\ 10^{1.076}\right] = [8.91,\ 11.91].$$

The parameter α represents the scaling exponent of brain size with respect to body size, while k is a proportionality constant representing the estimated brain size (in grams) for a mammal of unit body weight. The relatively narrow intervals indicate a high degree of precision in estimating the scaling relationship across species, reinforcing the empirical stability of the allometric law for mammalian brain–body evolution.

Using the estimated regression equation

$$\log(\widehat{\text{Brain}}) = 1.013 + 0.719\ \log(\text{Body}),$$

(under the assumption that the distribution of $\log(\varepsilon)$ is normal with mean 0) the corresponding 95% prediction interval for the brain size of a mammal with body weight x

(after back-transformation) becomes:

$$\left[10^{1.013+0.719\log x-0.5\sqrt{1.0104+0.00659(\log x-0.925)^2}},\ 10^{1.013+0.719\log x+0.5\sqrt{1.0104+0.00659(\log x-0.925)^2}}\right].$$

For species whose body weights are close to the mean of the data ($\log x \approx 0.925$), the term $(\log x - 0.925)^2$ becomes small, resulting in a narrow interval. However, as x deviates from the mean, the interval widens, indicating greater prediction uncertainty. The widening of prediction intervals for large mammals may reflect evolutionary diversity: species such as elephants or whales have undergone distinct neurological and ecological adaptations that deviate from the average scaling pattern observed across smaller species. Thus, while the regression captures the overall evolutionary trend in mammalian brain–body scaling, it cannot precisely predict the brain size of every individual species.

Evolutionary Implications. This allometric scaling offers meaningful insight into the evolutionary constraints acting on mammalian physiology. The sublinear exponent ($\alpha < 1$) implies diminishing returns in brain growth relative to body mass, suggesting that energetic and developmental costs limit the expansion of brain tissue in larger species. Conversely, deviations from this general trend may highlight species that have undergone selective pressures favoring greater neural complexity or cognitive ability. Thus, while the relationship is primarily physiological, it also encodes deep evolutionary trade-offs between intelligence, metabolic demand, and reproductive strategy.

Scope of Inference. The findings apply broadly to mammals as a group but must be interpreted with caution when comparing across distant taxonomic or ecological boundaries. Environmental adaptations, diet, and social behavior—all of which can influence brain development—are not captured in this dataset. Therefore, the inferences are most reliable at the macroevolutionary scale, illustrating general trends rather than species-specific predictions.

In conclusion, the analysis reinforces that brain size in mammals follows predictable scaling laws shaped by shared biological constraints, providing a quantitative framework for understanding one of evolution's most remarkable developments—the growth and diversification of the mammalian brain.

8 Appendix A: Dataset of Mammalian Characteristics

The dataset used in this study contains information on 96 species of mammals, including brain weight (in grams), body weight (in kilograms), gestation period (in days), and litter size. Due to the size of the dataset, the full table is provided here for reference. Each row corresponds to a single species.

Species	Brain	Body	Gestation	Litter
Aardvark	9.6	2.2	31	5
Acouchis	9.9	0.78	98	1.2
African elephant	4480	2800	655	1
Agoutis	20.3	2.8	104	1.3
Axis deer	219	89	218	1
Badger	53	6	60	2.2
Barbary sheep	210	66	158	1.2
Barking deer	124	16	183	1.1
Bat-eared fox	28.5	3.2	65	4
Beaked whale	500	250	240	1.8
Beaver	45	25	128	4
Black buck antilope	200	39	180	1
Bush baby	9.9	0.7	135	1
Canadian beaver	40	20	128	2.9
Capybara	76	30	123	3
Caribou	288	110	225	1
Cattle	456	520	280	1
Chimpanzee	360	45	230	1
Chinchilla	5.25	0.43	110	2
Deer mouse I	0.68	0.027	23	3.7
Deer mouse II	0.63	0.026	23	5
Deer mouse III	0.52	0.017	24	5
Deer mouse IV	0.69	0.024	24	5
Dog	70.2	8.5	63	4
Dolphin	1600	160	360	1
Domestic cat	28.4	2.5	63	4
Domestic goat	106	30	151	2

Domestic pig	180	190	115	8
Domestic sheep	125	49	150	2.4
Duikers	93	13	120	1
Eland	480	560	255	1
Elephant shrew I	1.14	0.049	51	1.5
Elephant shrew II	1.37	0.064	46	1.5
Elk	365	120	235	1
Fallow deer	223	80	240	1
Flying squirrel	1.89	0.052	40	3.1
Fur seal	355	250	254	1
Gentle lemur	7.8	0.22	145	2
Gorilla	406	140	265	1
Gray fox	37.3	3.8	63	3.7
Grizzly bear	400	250	219	2.3
Guinea pig	4.28	0.97	67	2.6
Hamadryas baboon	183	21	180	1
Hamster I	0.67	0.036	21	4.6
Hamster II	1.12	0.13	16	6.3
Harp seal	442	110	240	1
Hedgehog	3.5	0.93	34	4.6
Hippopotamus	590	1400	240	1
Hopping mouse	1.18	0.15	27	5.6
Horse	712	480	330	1
House mouse	0.45	0.024	19	5
Howler monkey	54	7.7	139	1
Human being	1300	65	270	1
Hyrax	20.5	3.8	225	2.4
Jack rabbit	13.3	2.9	41	2.5
Kinkajou	31.2	2	77	1.1
Leaf monkey	65.5	5.8	168	1
Lemur	22	2.1	135	1
Leopard	157	46	92	2.5
Lion	260	180	108	3
Llama	225	93	330	1
Long-nose armadillo	12	3.7	120	4

Lynx	75	12	60	2.5
Nutria	23	5	132	5.5
Orangutan	343	37	270	1
Porcupine I	37	11	112	1.2
Porcupine II	37	14	112	1.2
Porcupine III	24	6.6	113	1
Porpoise	537	56	270	1
Pygmy gerbil	1.04	0.065	21	4
Pygmy hippopotamus	260	150	205	1
Quokka	17.5	3.5	26	1
Raccoon	41.6	5.3	63	3.5
Rat I	0.72	0.05	23	7.3
Rat II	2.38	0.34	21	8
Red deer	435	200	255	1
Red fox	48	6	52	4
Rhesus monkey I	84.6	6	175	1
Rhesus monkey II	107	8.7	165	1.1
Ring-tail monkey	73	3.7	180	1
Sambar	383	120	246	1.1
Sea lion	363	100	343	1
Slow loris	12.8	1.2	90	1.2
Spider monkey I	114	9.1	140	1
Spider monkey II	109	7.7	140	1
Tapir	250	230	390	1
Tiger	302	210	104	3
Tree shrew	3.15	0.15	46	3
Tree squirrel	6.23	0.33	38	3
Vervet guenon	67	4.6	195	1
Vicuna	198	45	300	1.1
Weddell seal	550	400	310	1
Western baboon	179	32	180	1
White-handed gibbon	102	5.5	210	1
Wild boar	185	150	120	4
Yak	334	250	255	1

Table 8.1: Dataset of Mammalian Characteristics

9 Appendix B: Code and Computational Framework

All the codes and a CSV version of the dataset are provided here.

10 References

1. https://new.censusatschool.org.nz/resource/multivariate-data-sets/

2. https://kenbenoit.net/assets/courses/me104/logmodels2.pdf